UNIT 1 · TOPIC 6(B)

Threshing Equipment

Syllabus topic: Equipment for harvesting, threshing, mowing, chaff cutting and baling

Threshing

Threshing detaches grain from the ear head/panicle/pod — mostly by **impact (≈80%)**, plus rubbing and squeezing between the cylinder and concave.

Parts of a complete thresher, in order of crop flow

1. Feeding unit
2. Threshing cylinder & concave
3. Separating unit (straw walkers/louvers)
4. Cleaning unit (aspirator blower + oscillating sieves)
5. Grain elevator/bagging

Cylinder types (Kepner §17.4, Indian classification)

Type	Features / crops
Spike (peg) tooth	Spikes on drum vs. spikes on concave; vigorous — wheat (desi threshers); breaks straw into bhusa
Rasp bar	Corrugated bars; most universal — combines; gentle on straw, tolerates green weeds
Angle bar	Rubber-faced helical bars — clover, alfalfa (minimal seed damage)
Wire loop	Loops on drum — paddy, ragi (drummy/pedal threshers)
Hammer mill type	Beaters — hard-to-thresh crops; makes fine bhusa (e.g., syndicator)
Axial flow	Crop moves axially around the cylinder (spiral) — paddy thresher (IRRI design); high capacity, less grain damage

Settings

Crop	Cylinder peripheral speed (thumb rule)
Wheat	25–30 m/s (5000–6000 fpm — Kepner)
Paddy (wire loop)	12–18 m/s
Maize sheller	9–12 m/s

- Peripheral speed governs threshing more than drum diameter — a given peripheral speed threshes similarly whatever the drum size.
- Concave clearance decreases from the front (entry) to the rear. Smaller clearance and higher speed thresh better but break more grain and straw.
- Threshing effectiveness **rises** with cylinder speed, **falls** with higher feed rate, and **rises** with lower moisture.

Losses

Unthreshed (visible) loss, blown/sieve loss, spilled loss, broken (invisible/quality) loss.

Safety

- Feeding-chute length per **IS 9020** (safe feeding).
- **Dangerous Machines (Regulation) Act 1983.**

Practice MCQs

Q50. Threshing action in conventional threshers is mostly due to:

(a) Rubbing only (b) Impact
(c) Squeezing (d) Centrifugal force

Q51. The cylinder type predominant in modern grain combines is:

(a) Spike tooth (b) Rasp bar
(c) Wire loop (d) Hammer mill

Q52. Recommended cylinder peripheral speed for threshing wheat is about:

(a) 5–10 m/s (b) 12–15 m/s
(c) 25–30 m/s (d) 40–50 m/s

Q53. Cylinder–concave clearance in a thresher/combine is kept:

(a) Uniform (b) More at entry, less at exit
(c) Less at entry, more at exit (d) Zero at entry

Q54. In an axial-flow paddy thresher, the crop:

(a) Passes straight through (b) Moves spirally along the cylinder axis
(c) Falls by gravity only (d) Is threshed by rolls

Q55. Safe feeding-chute dimensions of power threshers are specified in:

(a) IS 7898 (b) IS 9020
(c) IS 4468 (d) IS 6024

Answers

Q50 — (b) Impact

Q51 — (b) Rasp bar

Q52 — (c) 25–30 m/s — *Kepner §17: 5000–6000 fpm ≈ 25–30 m/s for wheat.*

Q53 — (b) More at entry, less at exit — *A converging clearance grips the crop, then threshes it progressively without excess breakage.*

Q54 — (b) Moves spirally along the cylinder axis

Q55 — (b) IS 9020